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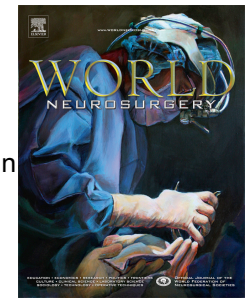

Analysis of Concussions Reported in American Professional Football Injury Reports in the 2019 through 2023 Regular Seasons

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PII: S1878-8750(24)00780-0

DOI: https://doi.org/10.1016/j.wneu.2024.05.026

Reference: WNEU 22418

To appear in: *World Neurosurgery*

Received Date: 2 May 2024

Accepted Date: 6 May 2024

Please cite this article as: O'Malley Jr GR, Sarwar SA, Posner KM, Polavarapu H, Adams D, Cassimatis ND, Munier S, Hundal JS, Patel NV, Analysis of Concussions Reported in American Professional Football Injury Reports in the 2019 through 2023 Regular Seasons, *World Neurosurgery* (2024), doi: https://doi.org/10.1016/j.wneu.2024.05.026.

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Keywords: Concussions, Football, Neurotrauma, Traumatic Brain Injury

Short Title: Treds in pro-football Concussions

Declarations of interest: None.

Funding: This study received no financial support from any institution or organization of any kind.

Acknowledgements: We would like to thank Drew Sturgill, Omid Tabatabaee, and Timothy Blajda for their assistance in data collection.

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Abstract

Objective

The National Football League (NFL) has seen increasing scrutiny regarding its handling of concussions, especially following an on-field incident involving Miami Dolphins' (™) Quarterback Tua Tagovailoa in the 2022 Season. The authors hoped to elucidate recent trends in the diagnosis and management of concussions over the course of five NFL seasons across 2019-2023.

Methods

The authors queried NFL injury reports from the 2019 through 2023 database recording players listed with Concussions. Weeks missed were calculated using NFL game logs. Player's concussions that did not occur in games, complicated by other injuries, or roster status were excluded.

Results

Searches of NFL injury reports resulted in the identification of 664 of 692 (96%) concussions that occurred in regular season games across the 2019-2023 seasons. Over the course of these five seasons 31% of players returned without missing a game, 39% of players missed one game, and 30% of players missed two or games. No significant difference in the number of concussions per game or weeks missed was observed across the seasons observed. Players with concussions on teams that made the playoffs saw less weeks missed than those on non-playoff teams (0.86 v. 1.37, p=0.002).

Conclusions

Since the start of the 2021 NFL season, an increasing incidence of concussions has been noted, yet there was no change observed in the number of weeks missed following concussions. Trends in the rates of concussions across seasons remain largely stable, despite increased scrutiny over concussions in the sport.

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Introduction

Neurotrauma in American football and particularly the National Football League (NFL) continues to be a major topic of discussion within the mainstream media and healthcare community. In recent years, there has been significant attention paid to concussions and their subsequent impact on the long term health of players. A concussion is a form of traumatic brain injury that results from a violent jolt or blow to the head that disrupts normal brain function[1]. This brain injury can result in symptoms such as headaches, dizziness and difficulty concentrating that usually improve within days to weeks, but can sometimes persist long term[2,3].

This issue came under renewed scrutiny after the Miami Dolphins' quarterback, Tua Tagovailoa, sustained a significant concussive injury against the Cincinnati Bengals during the 2022 season[4]. In the third week of the 2022 NFL season, Miami Dolphins' quarterback Tua Tagovailoa experienced a concerning incident. After a hit that caused his head to strike the ground, he appeared to have a concussion but cleared the concussion protocol and returned to play in the same game, despite exhibiting a disoriented gait. The following week, he endured a severe impact to the head, resulting in another suspected head injury. Post-impact, he appeared to have a fencing response position and was taken off the field on a backboard. This sequence of events led to intense scrutiny and criticism of the NFL's concussion protocol and raised concerns about long-term player safety.

The public outcry following Tua Tagovailoa's concussion led to an update in the NFL's concussion protocol, which became known as the "Tua Rule", and was jointly agreed upon by the NFL and the NFL Players Association (NFLPA)[4,5]. Under this updated protocol, ataxia joined the list of other symptoms, including but not limited to unconsciousness, confusion, or amnesia, that would mandate the immediate removal of a player from the game. Additionally, the updated protocol added certified athletic trainer (ATC) spotters to games in addition to the already required booth Unaffiliated Neurotrauma Consultant (UNC) spotters to specifically scan for signs of ataxia or other neurological symptoms in the name of player safety[4,5].

The Tua Rule is one example of how the NFL is updating its protocols to align with modern medical information and concerns. Emerging research shows possible links between repetitive concussive events and neurodegenerative diseases, including but not limited to Chronic Traumatic

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Encephalopathy (CTE), and Alzheimer's disease. Therefore, implementing and refining additional protocols may prove critical for long-term player safety[6,7].

As more understanding is gained regarding the biomechanics, pathophysiology, and long-term impacts of concussions and neurotrauma, there is an ever-present need to adapt current practices in the sport to provide the safest possible environment to players. The primary objective of this study is to examine trends in rates of concussion that occurred within regular season NFL games and time to return to play in the 2019 through 2023 NFL regular seasons. The secondary objectives of this study are to elucidate trends in concussions and their impacts regarding playing positions and team performance.

Methods

Study Design

This is an observational study examining trends surrounding the identification and return to play of NFL players who sustained concussions in the 2019 through 2023 NFL seasons using publicly available NFL injury reports. Approval from the local institutional review board was not needed, as this study did not meet definitions of human subject research.

Data Collection

The authors queried official NFL injury reports from the 2019-2023 seasons recording players who were listed with concussions. Players who underwent concussion evaluations but did not appear on injury reports in the week following the game were not recorded as this information is not publicly available. Weeks missed was calculated from participation in games and sourced from NFL player participation game logs. Player position and team information was extracted from NFL player profiles. The team for which the player was rostered on at the time of the concussion was recorded. The date on which the game was played was obtained using information from Pro Football Reference (Sports Reference, LLC). Playoff status was collected from the final NFL standings of the 2019-2023 seasons.

Return to Play Data Curation

Multiple factors can affect players' return to play after concussions, such as on-field performance, orthopedic injury, and roster status among others. To minimize confounding variables, the authors

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excluded players from the return to play analysis whose concussions were complicated by other injuries, roster status changes, or concussions that occurred at the end of seasons. Additionally, players who sustained concussions that occurred in non-football incidents were excluded.

Statistical Analysis

Data are presented as the percentages, mean, and standard deviation. Means and percentages were compared by Z-Scores. Statistical analysis was completed using R Software (including the packages: *tidyverse, skimr and FactoMineR* packages) with an alpha of 0.05.

Results

Concussion Rates

The concussions and their rates by season are described in Table 1. The number of concussions occurring in regular season games identified in this search over the 5 seasons were as follows: 2019=132, 2020=125, 2021=119, 2022=141, and 2023=147. Official NFL injury records report 692 concussions over this time period[8]. This study identified 664 of these 692 or 96% of all officially recorded in regular season game concussions.

The search of injury reports over the course of 2019 to 2023 resulted in the identification of concussions to 561 individual players. Over these five seasons, 473 players sustained one concussion, 74 players sustained two concussions in the same season, 13 players sustained three concussions and one player sustained four concussions. The breakdown by year for players who sustained 2 or more concussions in the same season is as follows: 2019=7, 2020=1, 2021=6, 2022=3, and 2023=2. The most concussions sustained by one individual player over the course of one regular season was three. This occurred during the 2021 season by a cornerback.

The number of concussions per game decreased from 2019 (0.55) to 2021 (0.47) but increased in the 2022 (0.54) and 2023 (0.57) seasons. The average number of concussions per week by season was as follows: 2019= 8.25, 2020= 7.44, 2021= 7.31, 2022= 8.38, 2023= 8.63. There was no significant variation in the number of concussions per week across seasons (F=0.568, p=0.6836). The distribution of concussions by week is graphically depicted in Figure 1.

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Concussions were further studied by the day of the game: Thursday=41, Saturday/Sunday=576, Monday=47. This distribution was separated into two groups: short rest between games (Thursday) and fullrest between games (Saturday/Sunday, Monday). In terms of concussions per game, games played after full rest (n=1270) saw 0.491 concussions per game, while games after a short rest period (n=90) saw 0.456 concussions per game (z=-0.64, p=0.5210).

The total number of concussions sustained per team over the course of a season ranged from 1-14. The mean number of concussions per season per team was as follows: 2019=4.26, 2020=3.79, 2021=3.72, 2022=4.27, 2023=4.74 (F=1.121, p=0.3486). The number of concussions each season and average over the course of the five seasons analyzed is displayed in Table 1. Over the course of 2019 to 2023 there were 68 playoff teams and 92 non-playoff teams that recorded 241 concussions and 423 concussions, respectively.

Cornerbacks were the most frequent position reported to sustain concussions over the course of the 2019-2023 NFL seasons with 119 concussions observed. As expected, the least frequent were kickers and punters with one concussion observed for each. Cornerbacks were followed by wide receivers with 116, and then offensive linemen, linebackers, and safeties with 94, 67, and 67, concussions respectively. The distribution of concussions correlating to positions that players play on the field is described in Figure 2.

Return to Play Analysis

For return to play following concussion analysis, the authors excluded concussions sustained at the same time as other injuries (n=11), players that were cut or traded following the concussion before returning to play (n=5), or concussions that occurred in the last two weeks of the season (n=47). The mean number of weeks missed following concussions by season was as follows: 2019=1.51, 2020=1.32, 2021=1.48, 2022=1.61, 2023=1.23. No significant difference was observed in regard to the mean number of weeks missed across the study period (F=0.70594, p=0.588073). The distribution of weeks missed per concussion per player is shown in Figure 3. Across the 2019 through 2023 seasons following concussions 31% of players returned without missing a game, 39% of players missed one game, and 30% of players missed two or games. The

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percentages of games missed following concussions are described on a season by season basis in Table 2.

When considering players that received multiple concussions that were included in the return to play analysis, over the five seasons (n=17), the mean number of weeks missed for the first concussion was 1.00, while the mean number of weeks missed for these players' second concussion of the season was 2.68 (p=0.000245). The mean amount of weeks between concussions was 6.6 with a range of 3 to 12 weeks for players who sustained at least two concussions.

A significant difference was observed between players' positions and weeks missed following a concussion (F=1.121, p=0.3486). The average number of weeks missed by each position as as follows: Cornerback=1.16, Defensive lineman=1.18, Fullback=1.00, Linebacker=1.58, Offensive Linemen=1.73, Quarterback=1.65, Running Back=1.44, Safety=0.92, Tight end=2.09, and Wide receiver=1.41. Return to play on a position by position basis is described in Table 2.

The range of weeks missed per concussion per team ranged from Philadelphia with 0.45 (n=22) to New England with 2.76 (n=25). It was found that teams that made the playoffs had significantly shorter return to play following concussions than teams that did not make the playoffs (1.19 v. 1.57, p=0.024664).

Before the Tua Tagovailoa injury (the 2019 season up through week 3 of the 2022 season), 350 concussions were observed, while after the Tua Tagovailoa injury 250 included concussions were observed (week 4 of 2022 season through 2023). Before the Tua Tagovailoa injury, the concussion rate per game was 0.429 while after the injury, the concussion rate per game increased to 0.521 ($p<0.01$). Before the Tagovailoa injury the mean weeks missed after a concussion was 1.51, while after the injury and implementation of the "Tua Rule" the mean weeks missed was 1.31 (p=0.24). Before and after the Tagovailoa incident there was no observed variation in percentages of players that returned without missing a game (32% v. 30%, p=0.5485), missing one game (37% v. 41%, p=0.35238), or missing more than one game (30% v. 29%, p=0.69654).

Discussion

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The results of this study report the findings of publicly available information regarding concussions in the 2019 through 2023 NFL regular seasons. These seasons are particularly notable as the length of the NFL regular season increased from 16 to 17 games following the 2020 season. This time period was further highlighted by the notoriety in which the management of concussions received following the concussion of Tua Tagovailoa and subsequent investigation into its management and rule changes that followed. Consistent with public reports, the total number of concussions was observed to increase in the study period from 2021 to 2023[8].

From publicly available injury records the authors were able to identify 664 of 692 or 96% of all officially recorded in regular season game concussions in 2019 to 2023. The findings of this study are largely consistent with prior studies using official NFL medical records investigating concussions and return to play[9]. From 2015 to 2020, Mack et al. similarly noted the highest number of concussions in the defensive secondary (safeties and cornerbacks), offensive line, and wide receivers. Additionally, Mack et al. also noted similar trends in regards to games missed by position.

However, compared to the period 2015-2020 which Mack et al. investigated, this study (2019-2023) does note a lower percentage of players clearing concussion protocol without missing a game (31.3% v. 40.7%, p=0.0001). While debate has existed in the media in regards to whether or not injuries increase on games played after short weeks of rest, i.e. Thursday night games, current literature has suggested that there is no correlation between Thursday night games and an increased rate of all cause injuries[10,11]. This study agrees with Mack et al. in that there was no observed rate of increased concussions on Thursday night games.

Other particularly notable findings include less games missed following concussions by teams which made the playoffs than did not. It has been observed in literature that NFL teams that perform better over the course of the season, typically endure less injuries and experience less weeks missed by their players[12]. The reason for this discrepancy is unclear. All players have to complete the same steps of return to play protocol before returning from a concussion. In theory these playoff teams may be “luckier” and sustain less severe concussions over the course of the regular season. Due to lack of access to the NFL's electronic medical record (EMR), the severity

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of concussions of individual players assessed in this study cannot be determined. It must also be considered if psychosocial factors, such as motivation could play a role in this discrepancy.

Additionally, the number of concussions sustained per game increased following the Tagovailoa injury compared to the time period before (0.521 v. 0.429, $p<0.01$). It is notable that the highest amount of concussions in a given week (17) was seen in the week of games immediately following the Tagovailoa Injury (Week 4, 2022). While there is an observed increase in concussions per game following the Tagovailoa injury, it is unlikely this difference is solely due to this event as major changes had taken place in the way in-game concussions were monitored in prior seasons.

Since 2012 there has been substantial growth in the number of individuals responsible for noticing possible concussion symptoms on NFL sidelines. In addition to emergency room physicians, team physicians and trainers, two athletic trainer spotters (ATS) and three unaffiliated neurotrauma consultants (UNC) are present at each game closely monitoring for hits that could lead to concussions, as well as symptoms[13]. Measures have been taken to increase the number of concussion evaluations with players with symptoms of conditions that can be associated with concussions, such as neck stingers[14]. As a result of the Tagovailoa injury, ATS and UNCs were instructed to look more closely at video reviews for helmet to helmet contact, as well as any motor or behavioral changes following collisions[15]. Under current NFL guidelines one of every three to four concussion evaluations resulted in one diagnosed concussion, as of 2023[16].

Notably, the changes the NFL made prior to and following Tua Tagovailoa's injury addressed those in regards to players with concussions with visible symptoms. It is important to note that the vast majority of concussions do not present with "gross motor instability" or the posturing that Tua showed after the Buffalo Bills and Cincinnati Bengals games respectively[17]. Most concussions will present with invisible symptoms of headache, nausea, a feeling of fog or slowing, phonophobia, among many other symptoms. These symptoms differ from other symptoms such as loss of balance, ataxia, vomiting and loss of consciousness and they can only be observed by the player that suffered concussions themselves. Furthermore, concussions may be associated with big hits, which concussions spotters are trained to look out for, but can also occur with a momentary

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bumping of heads. This bumping of heads can happen hundreds of times for players over the course of the game, especially for interior positional players such as linemen.

While concussion spotters play an important role in the prevention of concussed players to return to the game, they will not be perfect solutions to the problem due to their inability to see the invisible symptoms of concussions. The presence of invisible symptoms highlights the importance of players to self report their injuries. Literature has demonstrated college football player awareness of the symptoms to be high, as well as, knowledge of the possible long term complications of repetitive head injuries[18,19]. The NFL has stated that 43% of concussions in the 2023 NFL season were self-reported by players[16]. As recently as 2016 only 19% of concussions were self-reported by players[20]. There is currently no publicly available data in regard to the number of concussions identified by concussion spotters, officials, teammates, or coaches. Analysis of which individuals are reporting concussions on a season by season basis can allow for a better understanding of this relationship.

It is undoubtedly true that the number of reported concussions have increased overtime in professional football, it must be considered what percentage of the true number of concussions that these reported concussions make up. While it may lead to an apparent increase in the amount of concussions year to year, willingness of players to report symptoms can allow for proper healing of the brain from concussions and decrease the probability of long term complications. Further research is needed to evaluate professional football player's knowledge of the symptoms, attitudes to report concussions, and factors that would precipitate them being more willing to report symptoms.

Concerns over the use of publicly obtained data in the research of concussions has been previously reported in literature, mainly due to over-reporting of high profile players and incomplete data acquisition[21,22]. However, due to the necessary restrictions in accessing the health records of players, the majority of analysis of head injuries is performed as the results of public records searches. Public records searches in the future could benefit from standardized methodology and collaboration among researchers to verify injury incidence. Verified and accurate public records

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data could allow for third party authentication of trends and outcomes of professional sports injuries.

Limitations

This study is limited in that it does not use an official NFL database for concussions in regards to when they occurred, how long players were out for, nor the severity of the concussion. Additionally, an exact amount of days missed cannot be calculated as the missed games can only be measured in weeks due to games occurring every weekend. Concussions sustained in practices were not recorded in this study, thus a comparison between the rates of concussions sustained in games and practices could not be made. Further limitations are seen in return to play due to complicating factors such as other injuries, roster status, trades and overall playing ability. Moreover, all research into concussions is limited in that not all concussions can be identified and reported. Psychosocial factors likely play a large role in the reporting and return from concussions.

Conclusion

This study finds the number of total concussions has increased from the 2021 to 2023 seasons, yet little variation in the amount of games missed following a concussion. These increases could potentially be due to a combination of increased evaluations due to rule changes, and increased player awareness and willingness to self-report concussion symptoms, rather than the game becoming increasingly violent. More research is needed into NFL player willingness and knowledge to self report concussions. With not all concussions being visible to medical staff, concussion identification should be viewed as a team sport as player reporting is essential to mitigating potential missed concussions.

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Figure/Table Legend

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Figure 1. Concussions by week of the National Football League (NFL) Season over the course of 2019-2023. Week 4 of 2022 was the week of games following the Tua Tagovailoa incident and is designated with the color orange. Data from week 17 of the 2019 and 2020 seasons and week 18 of the 2021-2023 season were excluded due to poor data reporting as teams that did not make the playoffs did not publish injury reports.

Figure 2. Cumulative concussions by position over the course of the 2019-2023 National Football League (NFL) seasons as related to the players positions on a typical line of scrimmage. The color signifies the amount of concussions experienced by that position. Dark red signifies more than 90 concussions. Red signifies 60-89 concussions. Orange represents 40 to 59 concussions and yellow represents 0-39 concussions. Full Backs, Kickers, and Punters were excluded from the analysis due to only five, one, and one concussion being reported by these positions, respectively.

Figure 3. Weeks missed per player per concussion over the course of the 2019-2023 National Football League (NFL) seasons.

Table 1. Concussions by team by year. AFC denotes the American Football Conference Team in the city. NFC denotes the National Football Conference team in the city.

Table 2. The percentages of games missed following concussions are described on a season by season basis.

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Table 1. Concussions by team by year. AFC denotes the American Football Conference Team in the city. NFC denotes the National Football Conference team in the city.

Team	2019	2020	2021	2022	2023	Average	Standard Deviation
Arizona	2	2	4	1	0	1.80	1.48
Atlanta	8	5	3	2	7	5.00	2.55
Baltimore	2	3	4	3	6	3.60	1.52
Buffalo	6	4	3	3	4	4.00	1.22
Carolina	4	3	4	3	6	4.00	1.22
Chicago	5	3	4	6	6	4.80	1.30
Cincinnati	6	8	1	2	2	3.80	3.03
Cleveland	8	3	2	8	6	5.40	2.79
Dalas	1	3	1	2	2	1.80	0.84
Denver	1	3	1	2	2	1.80	0.84
Detroit	5	3	4	6	1	3.80	1.92
Green Bay	6	8	7	5	4	6.00	1.58
Houston	4	7	4	6	4	5.00	1.41
Indianapolis	5	4	5	8	9	6.20	2.17
Jacksonville	3	4	4	1	2	2.80	1.30
Kansas City	0	1	1	3	2	1.40	1.14
Los Angeles (AFC)	2	6	3	4	3	3.60	1.52
Los Angeles (NFC)	4	0	1	3	1	1.80	1.64
Las Vegas/Oakland	2	6	3	6	7	4.80	2.17
Miami	3	3	0	6	5	3.40	2.30
Minnesota	5	5	1	4	5	4.00	1.73
New England	3	2	4	9	8	5.20	3.11
New Orleans	3	6	1	5	7	4.40	2.41
New York (NFC)	14	3	2	6	5	6.00	4.74
New York (AFC)	2	6	8	3	6	5.00	2.45
Philadelphia	3	6	4	5	6	4.80	1.30
Pittsburgh	6	3	4	6	6	5.00	1.41
Seattle	1	2	5	2	3	2.60	1.52
San Francisco	3	4	8	3	5	4.60	2.07
Tampa Bay	1	3	4	3	6	3.40	1.82
Tennessee	5	2	8	5	7	5.40	2.30
Washington	8	3	6	6	5	5.60	1.82

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Table 2. The percentages of games missed following concussions are described on a season by season basis.

	Missed 0 Games		Missed 1 Game		Returned After 2 or More Games		
	Number	**Percent**	**Number**	**Percent**	**Number**	**Percent**	**Total**
2019	32	29.36%	40	36.70%	37	33.94%	109
2020	29	26.36%	45	40.91%	36	32.73%	110
2021	43	39.09%	40	36.36%	27	24.55%	110
2022	39	29.10%	54	40.30%	41	30.60%	134
2023	45	32.85%	55	40.15%	37	27.01%	137
Total	188	31.33%	234	39.00%	178	29.67%	600

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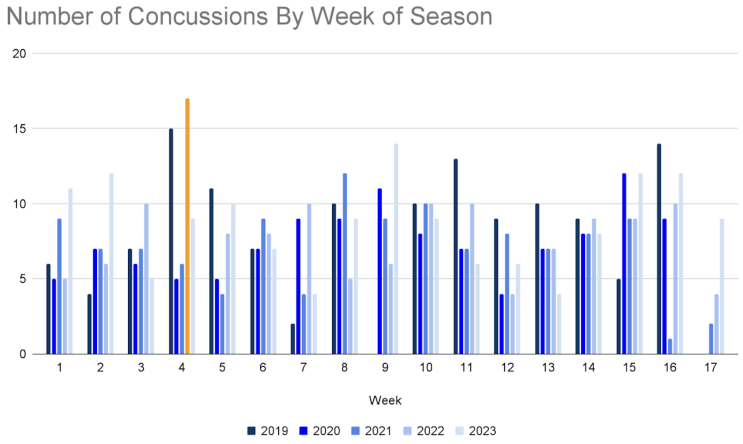

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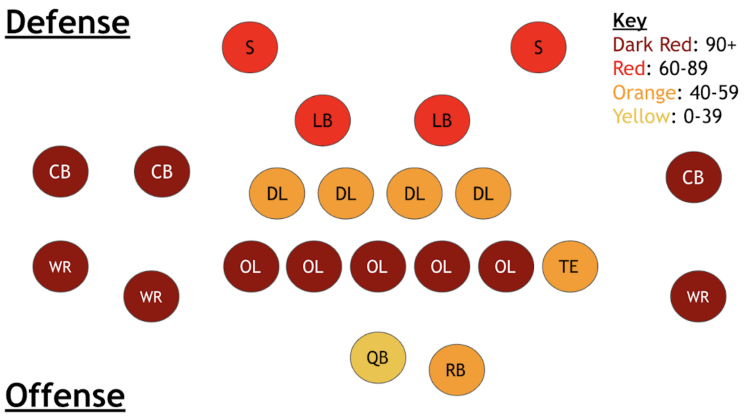

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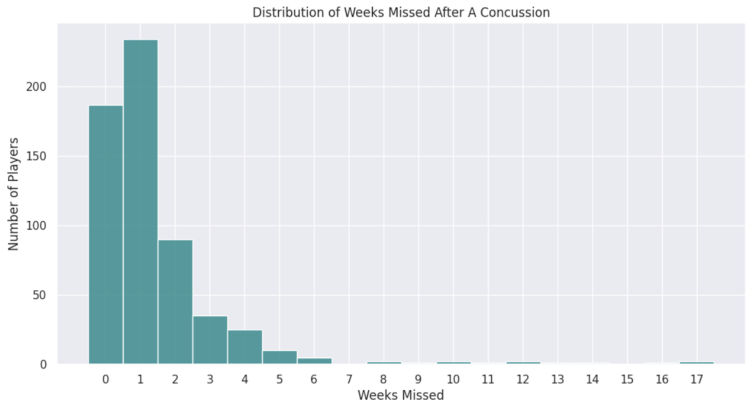

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NFL: National Football League
NFLPA: National Football League Players Association
ATC: Certified Athletic Trainer
UNC: Unaffiliated Neurotrauma Consultant
CTE: Chronic Traumatic Encephalopathy

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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